

The Exact KL Radius of Two Bernoulli Products on Three Bits

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Abstract

We give a computer-assisted proof candidate for the maximum forward Kullback–Leibler divergence from a distribution on three binary variables to a mixture of two Bernoulli products, equivalently the closure of $\text{RBM}(3,1)$. The exact value is $-\frac{3}{4}\log(2\sqrt{3}-3)$ nats; the only maximizing targets are the two uniform parity distributions. A three-multiplier log-supermodular certificate proves the lower bound. Reverse local stochastic channels reduce all targets to six support orbits, whose continuous simplices are covered by 16,600 exact witness pieces. A standard-library checker verifies 93,792 vertex conditions. We provide an executable rational target-to-witness interface and a conditioning corollary for mixtures of 2^{n-2} products on n bits. The constant was conjectured previously. Producer replay and internal model review are not unaffiliated verification or journal peer review.

Keywords: Kullback–Leibler divergence; latent class model; Bernoulli mixture; restricted Boltzmann machine; exact certificate.

1. Statement and answer

Write $X = \{0, 1\}^3$, ordered as 000, 001, 010, 011, 100, 101, 110, 111. Let

$$\mathcal{M} = \left\{ \lambda \bigotimes_{i=1}^3 \text{Ber}(u_i) + (1 - \lambda) \bigotimes_{i=1}^3 \text{Ber}(v_i) : \lambda, u_i, v_i \in [0, 1] \right\}.$$

This is the closure of binary $\text{RBM}(3,1)$. Define

$$\rho(p) = \min_{q \in \mathcal{M}} D(p||q), \quad D(p||q) = \sum_{x:p_x>0} p_x \log \frac{p_x}{q_x}.$$

A positive target entry divided by zero gives infinite divergence. The minimum exists: the parameter cube is compact, its image is compact, divergence is lower semicontinuous, and the uniform distribution supplies a finite competitor. Using the infimum over finite RBM parameters gives the same value, by approximation of the mixture parameters from the interior.

Theorem (computer-assisted).

$$\max_{p \in \Delta_7} \rho(p) = c = -\frac{3}{4}\log(2\sqrt{3}-3) = \frac{3}{4}\log\left(1 + \frac{2}{\sqrt{3}}\right).$$

Numerically, $c \approx 0.575738814443$ nats, or $c/\log 2 \approx 0.830615532444$ bits (12 decimal places). The only maximizing target distributions are

$$u_E = \frac{1}{4}(\delta_{000} + \delta_{011} + \delta_{101} + \delta_{110}), \quad u_O = \frac{1}{4}(\delta_{001} + \delta_{010} + \delta_{100} + \delta_{111}).$$

The proof has two parts. Section 2 gives an analytic lower bound and an exact parity projection. Sections 3–6 give the universal upper bound and classify all equality cases. The computer-assisted part is a finite convex-cover lemma, not a sample or an asymptotic numerical estimate.

2. Exact lower bound: the parity projection

Put $s = \sqrt{3}$,

$$a = \frac{2-s}{4}, \quad b = \frac{2s-3}{4}, \quad q^* = (\frac{1}{4}, a, a, b, a, b, b, 3a).$$

All entries are positive, and

$$q^* = \alpha \delta_{000} + (1-\alpha) \text{Ber}(t)^{\otimes 3}, \quad \alpha = \frac{3-s}{6}, \quad t = \frac{3-s}{2}. \quad (1)$$

Thus $q^* \in \mathcal{M}$. Direct substitution gives

$$D(u_E \| q^*) = -\frac{3}{4} \log(4b) = c. \quad (2)$$

It remains to exclude every better mixture; evaluating (2) alone would not do that.

2.1 Necessary supermodular inequalities

For coordinatewise meet and join, let A have one row

$$A_{xy} = e_{x \wedge y} + e_{x \vee y} - e_x - e_y$$

for each incomparable unordered pair x, y in the three-cube. There are nine rows. A positive probability table is log-supermodular in this orientation when $A \log q \geq 0$.

Every strictly positive-parameter mixture in $\mathcal{M}$ is log-supermodular after independent coordinate complements. To see this directly, complement coordinate i if necessary so that $u_i \leq v_i$. Factoring out the first product gives

$$\log q(x) = \text{a modular function of } x + \log(1 + \exp(d + \sum_i w_i x_i)), \quad w_i \geq 0.$$

The function $\phi(z) = \log(1 + e^z)$ has $\phi'' \geq 0$. Its supermodular difference is

$$\phi(z+r+t) + \phi(z) - \phi(z+r) - \phi(z+t) = \int_0^r \int_0^t \phi''(z+v+w) dw dv \geq 0$$

for $r, t \geq 0$. Hence all nine inequalities hold. Approximating boundary mixture parameters proves the corresponding closed polynomial inequalities there as well.

Complementing all three coordinates preserves the same cone, since it exchanges meet with join. The eight orientations therefore have four representatives. The even complements 0, 3, 5, 6 represent all four and preserve u_E . Only this necessary inclusion is used; sufficiency of the supermodularity conditions is not required by the proof.

2.2 A three-multiplier convex certificate

For arbitrary $z \in \mathbb{R}^8$, define

$$F(z) = \log \sum_x e^{z_x} - \sum_x (u_E)_x z_x.$$

This is convex, and $\nabla F(\log q^*) = q^* - u_E$. The three square-face rows with their upper vertex at 111 are

$$\begin{aligned} L_1 &= e_{001} + e_{111} - e_{011} - e_{101}, \\ L_2 &= e_{010} + e_{111} - e_{011} - e_{110}, \quad L_3 = e_{100} + e_{111} - e_{101} - e_{110}. \end{aligned}$$

They satisfy

$$L_i \log q^* = 0, \quad q^* - u_E = a(L_1 + L_2 + L_3), \quad a > 0. \quad (3)$$

Indeed, the active-face identities are $b^2 = 3a^2$. The remaining six supermodular inequalities are strict; they reduce to the positive quantities $b/4 - a^2$ and $3a/4 - ab$.

For any z in this cone, the convex supporting-plane inequality and (3) give

$$F(z) \geq F(\log q^*) + a \sum_{i=1}^3 L_i(z - \log q^*) \geq F(\log q^*).$$

Adding the constant $\sum_x (u_E)_x \log(u_E)_x$ proves $D(u_E \| q) \geq c$ throughout this cone. Applying an even complement gives the same bound in each of the other three orientations. Every mixture with interior parameters belongs to one of these cones. Approximation by interior mixture parameters extends the inequality to all of $\mathcal{M}$, including infinite-divergence cases. Together with (1)–(2), this proves

$$\rho(u_E) = \rho(u_O) = c. \quad (4)$$

For completeness, the four closest distributions to u_E are precisely the even translates $q^*(x \oplus f)$, $f \in \{0, 3, 5, 6\}$. For positive q , the preceding supporting-plane remainder equals $D(q^* \| q)$; thus equality within a cone forces $q = q^*$. A boundary q with a zero entry cannot give equality: along an interior approximation the same remainder tends to infinity because $q^* > 0$. Odd translates give the four projections for u_O .

The function `lower_certificate()` in `verify.py` checks (1), normalization, positivity, all nine polynomial inequalities, the three active faces, (3), and the objective identity exactly in $\mathbb{Q}(\sqrt{3})$.

3. Reduce every target to six support types

3.1 Local-channel monotonicity

A stochastic map on one binary coordinate takes a Bernoulli product to another Bernoulli product. It therefore preserves $\mathcal{M}$, as does a product of such maps. The log-sum inequality gives data processing,

$$D(Kp \| Kq) \leq D(p \| q).$$

Taking infima and using $K\mathcal{M} \subseteq \mathcal{M}$ gives

$$\rho(Kp) \leq \rho(p). \quad (5)$$

Coordinate permutations and complements preserve ρ exactly, since their inverses also preserve the model.

3.2 Removing an entry by reversing a channel

Fix one coordinate, with two unnormalized slices $u_y = p_{a,y}$ and $v_y = p_{b,y}$. Suppose both slices have nonempty support and

$$\text{supp}(u) \subseteq \text{supp}(v).$$

Define

$$\tau = \min_{y:u_y>0} \frac{v_y}{u_y} > 0, \quad r_{a,y} = (1 + \tau)u_y, \quad r_{b,y} = v_y - \tau u_y. \quad (6)$$

The table r is nonnegative and has total mass one. It has no new positive entry and has at least one fewer positive entry. Moreover, $p = Kr$, where

$$K(a | a) = \frac{1}{1 + \tau}, \quad K(b | a) = \frac{\tau}{1 + \tau}, \quad K(b | b) = 1. \quad (7)$$

Consequently $\rho(p) \leq \rho(r)$ by (5). Comparable slices in the other direction are treated by swapping a, b .

If a coordinate has only one nonempty slice, the distribution belongs to $\mathcal{M}$: any two-bit distribution is a mixture of two products by conditioning on either of its bits. Otherwise repeat (6) whenever possible. In at most seven steps, the process reaches either that zero-distance case or a support whose two projected slices are incomparable in each coordinate.

3.3 Exhaustive support classification

For a nonempty support $S \subseteq \{0, \dots, 7\}$ and a bit $b \in \{1, 2, 4\}$, form

$$S_0(b) = \{x \in S : x \& b = 0\}, \quad S_1(b) = \{x \oplus b : x \in S, x \& b \neq 0\}.$$

Retain S if neither of these two sets contains the other, for all three bits. Exhaustively applying this test to all 255 nonempty supports retains 50. Their orbits under the 48 coordinate permutations and complements have exactly the following representatives:

Representative, in binary-index notation	Orbit size	Cover pieces	Vertex checks
0, 7	4	2	4
0, 3, 5	8	18	54
0, 1, 2, 7	24	112	448
0, 3, 5, 6	2	648	2,592
0, 1, 2, 4, 7	8	4,226	21,130
0, 1, 2, 5, 6, 7	4	11,594	69,564
Total	50	16,600	93,792

The support enumeration is separately implemented in `support_certificate()`, without importing the discovery code. All weights on these supports remain arbitrary real probabilities. Enumerating supports does not discretize those weights; the next section covers their entire simplices. There are 52 prescribed roots in total and maximum split depth 20; both statistics are reconstructed in the replay receipt.

4. The finite convex-cover lemma

Lemma. For each of the six support simplices above, there is a finite simplicial cover such that every piece has a single explicit witness $q \in \mathcal{M}$ satisfying

$$D(v||q) \leq c \tag{8}$$

at every vertex v of that piece. Every verified vertex inequality is strict except when the vertex is exactly u_E or u_O ; those equalities use the corresponding translated witness from Section 2.

This is the computer-assisted lemma. Its exact data are the six JSON files in `certificates/`. Their hashes appear in `verification.json` and the bundle manifest. The checker proves the following finite assertions from those data; no discovery score is accepted as evidence.

4.1 The initial cover is explicit

For four of the support types, the root is simply the full probability simplex on the indicated vertices. For $S = \{0, 3, 5, 6\}$, take each of the 24 orders x_1, x_2, x_3, x_4 and the simplex with vertices

$$r_j = \frac{1}{j} \sum_{i=1}^j \delta_{x_i}, \quad j = 1, 2, 3, 4. \tag{9}$$

These simplices cover the parity simplex. Indeed, order the weights so that $p_{x_1} \geq \dots \geq p_{x_4}$, put $p_{x_5} = 0$, and use the exact convex decomposition

$$p = \sum_{j=1}^4 j(p_{x_j} - p_{x_{j+1}})r_j. \tag{10}$$

The coefficients are nonnegative and sum to one.

For $S = \{0, 1, 2, 4, 7\}$, use all orders of the odd-parity vertices and cone each simplex (9) from δ_0 . Formula (10), with an additional coefficient p_0 at δ_0 , proves coverage. All these root vertices are rational and are reconstructed by the checker.

4.2 Every subdivision preserves coverage

A split record names two distinct vertex positions i, j . Let $m = (v_i + v_j)/2$. Replace the parent simplex by two children, replacing v_i with m in one child and v_j with m in the other.

The children cover the parent: in barycentric coordinates λ , choose the first child if $\lambda_i \leq \lambda_j$, assigning weight $2\lambda_i$ to m and $\lambda_j - \lambda_i$ to v_j . The other case is symmetric. Conversely both children lie in the parent. All new vertices are rational.

Starting from the prescribed roots, the checker follows both children of every split, reconstructs their vertices with exact fractions, and requires every reached node to exist exactly once. It rejects invalid split edges, missing children, duplicate addresses, and unreachable extra nodes. Thus the final leaves form a genuine cover of every root simplex.

4.3 Witnesses belong to the requested model

There are only two witness formats.

1. A **star** witness specifies an atom f and means the translated distribution $q^*(x \oplus f)$ from (1). Membership is exact in $\mathbb{Q}(\sqrt{3})$.
2. A **mixture** witness gives seven integers and a positive common denominator. They specify $\lambda, u_1, u_2, u_3, v_1, v_2, v_3$. The checker verifies that all parameters are in $[0, 1]$, reconstructs the eight probabilities as exact rational numbers, and checks normalization and nonnegativity.

The search rounded mixture parameters, not probability-table entries. Hence membership does not rest on a tolerance test for tensor rank, a numerical decomposition, or approximate vanishing of a determinant.

4.4 The logarithm inequalities use exact arithmetic

For a rational $x > 0$, first write $x = 2^k u$, $1 \leq u < 2$. Set $y = (u - 1)/(u + 1)$, so $0 \leq y \leq 1/3$. Then

$$\log u = 2 \sum_{j=0}^{N-1} \frac{y^{2j+1}}{2j+1} + R_N, \quad 0 \leq R_N \leq \frac{9}{4(2N+1)3^{2N+1}}. \quad (11)$$

The tail estimate follows by bounding all remaining denominators by $2N + 1$ and summing a geometric series. The checker uses $N = 48$, fixed-point scale 2^{128} , and integer floor/ceiling operations after every multiplication and division. The same procedure bounds $\log 2$. Negative k reverses the endpoints of the multiplied interval. Thus every returned interval rigorously contains the logarithm.

For algebraic witnesses, an integer square root gives

$$\frac{L}{2^{128}} < \sqrt{3} < \frac{L+1}{2^{128}}, \quad L = \left\lfloor \sqrt{3} 2^{256} \right\rfloor.$$

The checker verifies both bounding square inequalities. Affine interval evaluation and monotonicity of the logarithm then enclose every star probability and the constant c .

For each non-equality vertex, the checker evaluates an upper bound on

$$\sum_x v_x \log v_x - \sum_x v_x \log q_x$$

and proves it is strictly below a lower bound on c . A zero target term is omitted; a positive target over a zero witness entry is rejected. No floating-point logarithm or floating-point comparison participates in these proof decisions.

At an equality vertex, the checker requires the exact uniform parity vector and a star atom of matching parity. The equality is then the exact identity (2), not a numerical tolerance exception. All other vertices require a strict inequality. The smallest certified strict vertex slack in this cover exceeds 2×10^{-6} nats. This is a margin for the finite vertex tests, not a uniform separation of all nonmaximizing targets from c .

4.5 Replay result

Run, using Python 3.10 or later with only its standard library:

```
python3 verify.py --out replay.json
python3 test_verify.py
```

The first command reconstructs the lower-bound algebra, all support orbits, all six covers, and all 93,792 vertex inequalities. The recorded local run returned **PASS**, with 16,600 leaves. There are 552 incidences of exact parity vertices across overlapping pieces; every other vertex test is strict. The second command passed 13 tests, including malformed-certificate rejection, an independent Decimal-logarithm diagnostic, and 765 exact rational channel-reconstruction checks.

The verifier imports no optimization package or discovery implementation. Numerical search suggested the witnesses; the exact replay proves the finite lemma. The remaining trusted base is the mathematical argument in this document, the checker implementation, Python’s standard-library integer/fraction semantics, and the execution platform. No claim of proof-assistant certification or independent specialist review is made.

5. Universal upper bound

Fix a support-simplex point p . By Section 4 it belongs to a leaf simplex with vertices $v_1, \dots, v_m$ and a common witness $q \in \mathcal{M}$. Write $p = \sum_j \lambda_j v_j$. For fixed q , divergence is convex in its first argument, since $t \log t$ is convex and the cross-entropy term is linear. Therefore

$$\rho(p) \leq D(p||q) \leq \sum_j \lambda_j D(v_j||q) \leq c. \quad (12)$$

This establishes the bound for every real weight vector on all six representative supports. Cube symmetries give the same conclusion on every irreducible support. Reversing the reductions in Section 3 and applying (5) gives $\rho(p) \leq c$ for every distribution on all eight cube vertices. Together with (4), this proves the exact global value.

Notice the quantifier: (12) covers every point in each simplex, not just its vertices as a finite list of target distributions. Convexity is precisely what turns the finite certificate into a universal upper bound.

6. Classification of all maximizing targets

First consider a target on an irreducible support. Every non-parity vertex of every leaf has a strict bound. If a target is not a leaf vertex, its barycentric representation uses at least two distinct vertices. On the finite-divergence simplex, $D(\cdot||q)$ is strictly convex: $\sum_x p_x \log p_x$ is strictly convex and the other term is linear. Thus (12) is strict for any such target. It follows that equality on irreducible supports is possible only at a uniform parity vector. Cube symmetries only exchange the two parity vectors.

Now take a target for which Section 3 made at least one strict support reduction. If its terminal target is not uniform parity, monotonicity already gives $\rho(p) < c$. Otherwise consider the last inverse channel K in (7), so its input is uniform parity u . Let $q_u^* > 0$ be a parity projection from Section 2. Then

$$\rho(Ku) \leq D(Ku||Kq_u^*) < D(u||q_u^*) = c. \quad (13)$$

The middle inequality is strict by the equality condition in the log-sum inequality. In each output fiber with changed bit b , the channel merges the two input values a, b with positive q_u^* -mass from both. Exactly one of the two parity target entries is positive. Their likelihood ratios relative to q_u^* are consequently different, while $\tau > 0$ ensures genuine merging. Earlier channels preserve the strict upper bound by (5).

Targets with a deterministic coordinate have zero error. Hence the only global maximizers are u_E and u_O , as claimed.

7. Constructive content and dimensional transfer

The upper bound supplies an explicit procedure, not just an existence argument. For a given target, apply (6), recording each channel. Transform the terminal support to a representative. Sort its weights to choose a root in (9), if needed. Descend the split tree using barycentric coordinates. Read the exact witness at its leaf, undo the cube symmetry, and apply the recorded channels. The resulting distribution remains a mixture of two products and has divergence at most c .

The command-line implementation `witness.py` accepts exact rational input probabilities. It verifies all six certificates before answering queries, performs the channel and barycentric calculations with fractions, and returns mixture weights and Bernoulli parameters in $\mathbb{Q}(\sqrt{3})$, together with the reduction, symmetry and leaf trace. It returns a feasible approximation certified to have divergence at most c , **not** a generally optimal projection or an algorithm for training an RBM. The written procedure applies to arbitrary real targets; the executable interface is restricted to rational targets. Bit-coordinate arrays in this interface are explicitly least-significant-bit first.

7.1 Worked channel and tree query

Take $p = (3, 1, 0, 4, 0, 4, 3, 1)/16$. For the least significant bit, the slices with bit values 0 and 1 have comparable supports, and (6) gives $\tau = 1/3$. The reduced target is exactly u_E . The deterministic tie rule orders its support as $(0, 3, 5, 6)$, selecting root 0 with barycentric coordinates $(0, 0, 0, 1)$. Descending the recorded tree locates a leaf containing that parity vertex and its matching star witness. Undoing the recorded channel produces a two-product mixture for p . The strict data-processing argument in (13) shows its error is below c . The exact leaf address, final parameters and barycentric weights are retained in `examples/channel-witness.json`.

```
python3 witness.py 3/16 1/16 0 1/4 0 1/4 3/16 1/16 --out query.json
python3 test_witness.py
```

7.2 A conditioning transfer theorem

For positive integers n, k , let $\mathcal{M}_{n,k}$ denote mixtures of at most k Bernoulli products on n bits, with closed parameters, and let $R(n, k) = \sup_p \min_{q \in \mathcal{M}_{n,k}} D(p||q)$.

Proposition (conditioning lift). If $n \geq m \geq 1$, then

$$R(n, 2^{n-m}k) \leq R(m, k). \quad (14)$$

Proof. Write a target as $p(z, y) = p_Z(z)p(y | z)$, with z consisting of $n - m$ fixed coordinates. Ignore zero-mass contexts. For each other context choose a k -product approximation q_z to its conditional law with error at most $R(m, k)$. The distribution $q(z, y) = p_Z(z)q_z(y)$ is a mixture of at most $2^{n-m}k$ products: append the deterministic coordinates z to every component of q_z , and multiply its weight by $p_Z(z)$. The supports of the contexts are disjoint, so cancellation of the common context weights gives

$$D(p||q) = \sum_{z:p_Z(z)>0} p_Z(z)D(p(\cdot | z)||q_z) \leq R(m, k).$$

Taking the minimum and supremum proves (14). This is the standard disjoint-support mixture principle, written here explicitly; no novelty is claimed for the lifting mechanism.

Corollary. The three-bit theorem implies, for every $n \geq 3$ and every $k \geq 2^{n-2}$,

$$R(n, k) \leq c = -\frac{3}{4} \log(2\sqrt{3} - 3). \quad (15)$$

In bits this upper bound is $c/\log 2 = 0.830615532444310205 \dots$. It improves the elementary one-bit upper bound at $k = 2^{n-2}$: conditioning on $n-2$ bits and approximating each remaining two-bit table by its independent marginals gives error $I(X; Y) \leq \log 2$. The mechanism is established in the disjoint-support mixture literature; the quantitative improvement is a consequence of the candidate three-bit constant. Sharpness for $n > 3$, maximizers in higher dimensions, stronger bounds at other component budgets, and consequences for general multi-hidden-unit RBMs are **not** established. In particular, product mixtures and RBMs must not be interchanged for these larger parameter values.

Oracle.approximate implements (14) for $m = 3, k = 2$. Its tests cover 765 rational three-bit targets over every nonempty support, both parity targets, the worked channel example, and 50 four-/five-bit conditioning examples. These finite tests diagnose the implementation; the all-dimensional claim is the analytic argument above.

The retained discovery scripts are optional. They are not imported by the checker and need NumPy/SciPy only if the search is rerun. A numerical run need not regenerate byte-identical witnesses on a different optimizer version; the frozen exact certificate, not deterministic discovery, is the proof object.

No boundary-MLE stratum classification, floating-point EM fit, approximate global optimizer, unproved entropy inequality, or symmetry average of mixtures is needed in the final proof.

8. Prior art, scope, and review boundary

The problem is the frozen AIM/UnsolvedMath item AIM-PROBABILITY-0002, not the neighboring RBM(3,2)/three-component-mixture problem. Montúfar’s 2018 review, Section 9, item 10, already records the proposed value, attributed to discussions with Johannes Rauh. Its formulation and numerical constant are not claimed as discoveries here. The current full-scope contribution proposed for review is the channel reduction plus the exact convex cover, together with the short parity certificate and equality argument.

The necessary supermodularity description is consistent with Allman–Rhodes–Sturmfels–Zwiernik’s nonnegative-rank-two framework. Section 2 proves the inclusion actually needed, so the final argument does not silently rely on a positive-support characterization at zero entries. The exact boundary-MLE paper was useful during reconnaissance but is not a dependency of the final proof.

Alexandr and Hoşten (2025), Proposition 2, use strict convexity to localize divergence maximizers at vertices of logarithmic Voronoi polytopes for linear and toric models. The shared convexity principle is established material. Our cover consists instead of prescribed simplices with feasible common witnesses for a nonconvex two-product mixture model: a witness need not be the projection throughout its simplex. The new proof object does not reconstruct logarithmic Voronoi polytopes. Montúfar, Rauh and Ay (2013), Lemma 6, give the established disjoint-support decomposition underlying Section 7.2.

The bounded novelty searches found no source proving this same global equality. That does not establish priority, present-day openness, or absence of unpublished work. The accompanying novelty note records the search limits. Internal model-mediated review and producer replay are not independent mathematical review. The constructive interface and the conditioning corollary

demonstrate scoped reuse, not measured impact or a substantiated four-star assessment.

9. Proof-to-checker map and disclosure

Obligation	Exact executable check	Analytic bridge retained in the manuscript
Parity lower bound	<code>lower_certificate</code> : mixture identity, signs, nine inequalities, three multipliers	Necessary cone inclusion, convex supporting plane, boundary passage and uniqueness; Section 2
All targets reduce to six support orbits	<code>support_certificate</code> : all 255 supports, 50 irreducibles and six orbits	Reverse-channel formula, strict support decrease and data processing; Section 3
Continuous simplex coverage	<code>root_simplices</code> , <code>verify_cover</code> : roots, every branch, no orphan node	Ordered-weight root decomposition and midpoint coverage; Sections 4.1–4.2
Feasible common witnesses	<code>witness_logs</code> , <code>lower_certificate</code>	Product-mixture model definition; Sections 2 and 4.3
Strict vertex error bounds	<code>log_interval</code> , <code>vertex_slack</code> , exact parity exception	Log-series remainder and first-argument convexity; Sections 4.4–5
Only two maximizers	Strict vertex tests and exact parity recognition	Strict convexity and strict log-sum equality condition; Section 6
Constructive query and lift	<code>witness.py</code> , <code>test_witness.py</code>	Channel reversal and disjoint-context KL identity; Section 7

The checker is a standalone implementation separated from discovery within the same producer-coordinated workflow. The test suite is not a formal verification. Neither the lower-bound analytic bridges nor the all-dimensional quantifiers are reduced to checking finitely many random inputs.

The scholarly creator is Anonymous; Evidence Press is the publisher, not an inferred personal author. A human directed the research and supplied review; an OpenAI Codex research assistant drafted the mathematical argument, implemented search and verification, revised the manuscript and prepared the release. Producer-coordinated model reports provide internal editorial scrutiny only. No external funding or competing interests have been declared; no human/animal subjects or personal datasets are involved. These disclosures do not establish independent validation.

All original prose and mathematical certificate data are offered under CC0-1.0; original code is MIT. The package links third-party literature and code without relicensing them. The supplied review is not redistributed. The paper, source, exact certificates, executable witness interface, semantic controls and original response matrix are available together in the versioned evidence package. Optional discovery requires the separately listed numerical dependencies; all proof replay and query operations use Python’s standard library.

Sources

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The primary sources, GitHub implementation, and live Evidence Press context are identified in NOVELTY.md. Original screening excerpts remain in the preserved review-v1 archive and are not redistributed in this successor.

Traditional Chinese abstract

本文提出一項電腦輔助證明候選結果，求出三個二元變數的機率分布到兩個伯努利乘積分布混合模型的最大正向 KL 散度。其精確值為 $-\frac{3}{4}\log(2\sqrt{3}-3)$ ，單位為奈特；僅有兩個均勻奇偶分布達到最大值。證明結合反向局部隨機通道、對數超模性，以及可重播的精確凸覆蓋證書。另提供從有理目標分布產生可行近似的程式，以及透過條件分布得到的高維上界。這是尚未經外部同儕審查的匿名人工智慧輔助候選論文，並未宣稱獨立驗證或高維上界的銳利性。